# Min Woo Park, Ph.D.

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**SHORT BIO.** During my Ph.D., I focused on *causal decision-making* (NeurIPS-25, UAI-26a, ICLR-26) and *causal effect identification* (NeurIPS-24, UAI-26c). I investigated how to formally model diverse real-world problems within a causal framework and derive optimal decisions from such structured representations.

In particular, I developed methods for robust decision-making under uncertainty (NeurIPS-25), domain shifts (UAI-26a), and even counterfactual actions (ICLR-26).

## EDUCATION

**Graduate School of Data Science, Seoul National University** — Seoul, Republic of Korea
**Ph.D.** in Data Science — Sep. 2022 – Feb. 2027
Advisor: Prof. Sanghack Lee.
Thesis: *Generalizing Causal Decision-Making: Markov Equivalence, Transportability and Counterfactuals*.
Doctoral Qualifying Exam waived due to outstanding academic performance.

**Department of Mathematics and Statistics, Sejong University** — Seoul, Republic of Korea
**B.S.** in Applied statistics and Mathematics (Double major), *Cum Laude* — Mar. 2016 – Aug. 2022
Advisors: Prof. Ji Eun Lee and Jin Hee Yoon. — (2.5 years military service)

## RESEARCH EXPERIENCE

**Samsung Research** | AI Research Engineer — Starting in Mar. 2027
**Furiosa AI** | AI Research Engineer Internship — Jun. 2026 – Sep. 2026
**Causality Lab, Seoul National University** | Ph.D. Researcher — Jan. 2023 – Feb. 2027
**Functional Analysis Lab, Sejong University** | Undergraduate Researcher — Jul. 2021 – Dec. 2022
**ISDS Lab, Sejong University** | Undergraduate Researcher — Jan. 2021 – Dec. 2022

## PUBLICATIONS

* for joint authorship and † for corresponding author.

- ◇ Generalizing Causal Decision-Making: Markov Equivalence, Transportability and Counterfactuals.
  **Min Woo Park** and Sanghack Lee†.
  *Causality Lab, Seoul National University, Technical Report*, 2026.

- ◇ On Transportability for Structural Causal Bandits.
  **Min Woo Park** and Sanghack Lee†.
  *Uncertainty in Artificial Intelligence* (UAI), 2026.

- ◇ Breaking Bad: Component-wise Parent Deletion for Score-Based Causal Discovery.
  **Min Woo Park***, Taehui Yun*, Youngin Jang, Younseok Yeom, Jonghwan Kim, Jiyeon Kang, Songseong Kim, Hyemin Jung, Sangmin Lee, Jongseong Jang†, and Sanghack Lee†.
  *Uncertainty in Artificial Intelligence* (UAI), 2026.

- ◇ Canonical Domain Reduction for Partial Counterfactual Identification.
  Yesong Choe, Yeahoon Kwon*, **Min Woo Park***, and Sanghack Lee†.
  *Uncertainty in Artificial Intelligence* (UAI), 2026.

- ◇ Counterfactual Structural Causal Bandits.
  **Min Woo Park** and Sanghack Lee†.
  *International Conference on Learning Representations* (ICLR), 2026.

- ◇ Structural Causal Bandits under Markov Equivalence.
  **Min Woo Park**, Andy Arditi, Elias Bareinboim†, Sanghack Lee†.
  *Neural Information Processing Systems* (NeurIPS), 2025
  CausalAI Laboratory, Columbia University, Technical Report (R-122).

- ◇ Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference.
  Yonghan Jung†, **Min Woo Park**, and Sanghack Lee†.
  *Neural Information Processing Systems* (NeurIPS), 2024.

◇ Computation of the iterated Aluthge, Duggal, and Mean transforms.
**Min Woo Park**[*], and Ji Eun Lee[*†].
*Filomat*, 2023.

◇ Convergence of the iterated mean transforms of a $2 \times 2$ matrix
Ji Eun Lee[*†] and **Min Woo Park**[*].
under review.

## AWARDS AND HONORS

**NRF Ph.D. fellowship in science and engineering, 2026** | National Research Foundation of Korea
▷ *Active Learning-Based Action Space Optimization for Causal Multi-Armed Bandits under Incomplete Information*. Sep. 2026 – Aug. 2027

**UAI 2026 scholarship** Aug. 2026

**Graduate Student Instructor (GSI) scholarship** Spring 2024, Spring 2025, Spring 2026

**BK21 FOUR scholarship** Fall 2024, Fall 2025

**First place merit scholarship** | Sejong University Fall 2020, Fall 2021

**Fourth place award** | The 9th College of Natural Sciences Academic Conference, Sejong University
▷ *Mitigating Spurious Regression in Time-Series Data via Granger Causality*. Nov. 2021

**Third place award** | The 8th College of Natural Sciences Academic Conference, Sejong University
▷ *The Significance of the Euler's Number and Its Applications*. Nov. 2020

## PROJECTS

**Towards More Scalable Score-based Causal Discovery** | LG AI Research Sep. 2025 – Aug. 2026
▷ Leading a project on scalable score-based causal discovery algorithms.
- Developed a novel operator enabling score-based causal discovery algorithms to escape local optima.
- Investigating parallelization strategies for score-based causal discovery algorithms.

▷ Resulting paper accepted at *Uncertainty in Artificial Intelligence* (UAI-26b).
- Software available at `https://github.com/LGAI-Research/breaking-bad`.

▷ Korean patent application filed.

**Hyundai NGV Data Boot-Campus** | Hyundai Motor Company
▷ **2025.** Developed high-voltage battery performance prediction technology. Jul. 2025 – Sep. 2025

▷ **2024.** Developed an intelligent system for predicting vulnerable regions of automotive body panel stiffness using differential geometrybased automatic curvature analysis. Jul. 2024 – Sep. 2024

**Optimal Mixed Policies Completeness** | Oxford University Aug. 2023 – Feb. 2024
▷ Studied characterization of conditions under which an intervention effect of a variable is positive.

## TEACHING ASSISTANTS

**Data Science Project** Fall 2026
▷ Practical experience on end-to-end project execution.

**Causal Inference for Data Science** Spring 2024, Fall 2024, Spring 2025, Spring 2026
▷ Topics included Bayesian networks and graphical causal inference.

**Machine Learning and Deep Learning for Data Science** II Fall 2023
▷ Covered kernel methods, Gaussian processes, and related topics in machine learning.

**Basic Mathematics for Artificial Intelligence (Undergraduate)** Spring 2022
▷ Covered linear algebra, basic statistics, and probability theory.

## ACADEMIC SERVICES, PROFESSIONAL TALKS AND POSTERS

**Academic Services**

Journal Reviewer:

Journal of the American Statistical Association (JASA)

The Journal of Artificial Intelligence (AIJ)

Conference Reviewer: **2027** ICLR, **2026** NeurIPS, UAI, ICLR, ICML, **2025** ICML

Reviewed AI/ML lecture video content from LG AI Research for LG Aimers
Summer 2025, Winter 2025, Summer 2026

**Invited Talks**

Major Exploration for Sejong University Students | Seoul, Republic of Korea — Nov. 2026

SNU GSDS student research seminar | Seoul, Republic of Korea — Dec. 2024, Apr. 2025

**Poster Sessions**

UAI 2026 poster session | Amsterdam, Netherlands — Aug. 2026

ICLR 2026 poster session | Rio de Janeiro, Brazil — Apr. 2026

ExploreCSR poster session | Seoul, Republic of Korea — Dec. 2025

NeurIPS 2025 poster session | San Diego, USA — Dec. 2025

JKAIA 2025, NeurIPS preview session | Seoul, Republic of Korea — Nov. 2025

SNU GSDS workshop poster session | Jeju, Republic of Korea — Jul. 2025

NeurIPS 2024 poster session | Vancouver, Canada — Dec. 2024

## REFERENCE

**Sanghack Lee** | Graduate School of Data Science, Seoul National University — sanghack@snu.ac.kr
Associate professor

**Ji Eun Lee** | Department of Mathematics and Statistics, Sejong University — jieunlee7@sejong.ac.kr
Associate professor

**Jin Hee Yoon** | Dept. of AI and Information Technology, Sejong University — jin9135@sejong.ac.kr
Associate professor

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